

Practical exercises — ratio and proportion

A Cambridge insert: sharing in a ratio, scaling a recipe, and telling direct from inverse.

LESSON 167

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 12

LESSON CLOCK

8'

12'

12'

6'

2'

Ratios and their parts	8 min
Sharing in a ratio	12 min
The unitary method	12 min
Direct and inverse	6 min
Homework	2 min

LEARNING OBJECTIVES

- Simplify a ratio and share a quantity in a given ratio.
- Use the unitary method to scale a recipe or a price.
- Distinguish direct from inverse proportion.
- Use a scale on a map or a plan.

TEXTBOOK

National	pp. 507–510
Cambridge	Stage 8, pp. 118–125
Workbook	Exercise 12.2

01

Explanation

Ratios and their parts

A ratio compares quantities of the same kind. It simplifies like a fraction, by dividing every part by their common factor.

RATIO	SIMPLIFIED	PARTS IN ALL
12 : 18	2 : 3	5
30 : 45 : 75	2 : 3 : 5	10
0.5 : 2	1 : 4	5
$\frac{1}{2} : \frac{1}{3}$	3 : 2	5

BOTH QUANTITIES MUST BE IN THE SAME UNITS

50 cm : 2 m is not 50 : 2. Convert first: 50 : 200 = 1 : 4.

Sharing in a ratio

STEP			SHARING 120 IN THE RATIO 2:3		
add the parts			$2 + 3 = 5$		
find one part			$120 \div 5 = 24$		
multiply out			$2 \cdot 24 = 48, 3 \cdot 24 = 72$		
check			$48 + 72 = 120 \checkmark$		

AMOUNT		RATIO	SHARES
120		2 : 3	48, 72
210		3 : 4	90, 120
300		1 : 2 : 3	50, 100, 150

THE SHARES MUST REBUILD THE TOTAL

Adding them back is a two-second check that catches nearly every arithmetic slip in this topic.

The unitary method

Find the value of one unit, then multiply up.

PROBLEM	ONE UNIT	ANSWER
5 pens cost 15 000: 8 pens	3000	24 000
a recipe for 4 needs 600 g flour: for 6	150 g	900 g
3 workers lay 90 m in a day: 7 workers	30 m	210 m

SCALES ARE RATIOS TOO

A map at 1 : 100 000 shows 1 cm for every 1 km on the ground, so 7 cm between two towns means 7 km.
Read the scale as “one of these is that many of those”.

Direct and inverse

KIND	AS ONE GROWS	CONSTANT	EXAMPLE
direct	the other grows	$\frac{y}{x}$	cost of n identical pens
inverse	the other shrinks	xy	workers and the days a job takes

Example: 4 workers finish a job in 6 days; how long would 8 take? The product $4 \cdot 6 = 24$ worker-days is fixed, so 8 workers need 3 days.

NOT EVERYTHING THAT CHANGES TOGETHER IS PROPORTIONAL

Doubling the workers may not halve the time if they get in each other's way, and a child twice as old is not twice as tall. Proportion is a model, and it has to be checked against sense.

02

Worked examples

EXAMPLE 1 Share 120 in the ratio 2 : 3.

1

2 + 3 = 5 parts.

2

120 ÷ 5 = 24 for one part.

3

48 and 72.
Check: 48 + 72 = 120 ✓

ANSWER

48 and 72

EXAMPLE 2 A recipe for 4 people needs 600 g of flour. How much for 6?

① One person: $600 \div 4 = 150$ g.

② Six people: $150 \cdot 6$.

③ $= 900$ g

ANSWER

900 g

EXAMPLE 3 4 workers finish a job in 6 days. How long would 8 workers take?

① More workers, less time — inverse proportion.

② $4 \cdot 6 = 24$ worker-days.

③ $24 \div 8 = 3$

④ Three days.

ANSWER

3 days

03 Interactive model

Scale a real recipe from four people to the size of the class; the arithmetic is identical to the textbook exercise and nobody asks what it is for.

Direct, inverse, or neither?

12 : 18 simplifies to:

2 : 3

3 : 2

6 : 9

1 : 2

50 cm : 2 m is:

50 : 2

1 : 4

25 : 1

1 : 40

120 in the ratio *2 : 3* gives:

40, 80

48, 72

50, 70

60, 60

5 pens cost 15 000; 8 cost:

20 000

24 000

30 000

18 000

Cost against number of identical pens is:

direct

inverse

neither

constant

Workers against days for a fixed job is:

direct

inverse

neither

constant

4 workers take 6 days; 8 take:

12

3

2

4

A map at 1 : 100 000: 7 cm means:

700 m

7 km

70 km

7 m

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
Proportion	Proporsiya	Пропорция
To share	Taqsimlash	Разделить
Unitary method	Birlik usuli	Метод приведения к единице
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Scale	Masshtab	Масштаб
Equivalent ratios	Teng nisbatlar	Равные отношения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$30 : 45 : 75$ simplifies to:

$2 : 3 : 5$

$1 : 2 : 3$

$3 : 4 : 5$

$6 : 9 : 15$

Sharing needs first:

the total

the sum of the parts

one share

the ratio reversed

210 in the ratio $3 : 4$ gives:

$90, 120$

$100, 110$

$70, 140$

$105, 105$

The unitary method finds:

the total

the value of one unit

the ratio

the difference

In inverse proportion the constant is:

$$\frac{y}{x}$$

xy

$x + y$

$x - y$

A recipe for 4 scaled to 6 multiplies amounts by:

1.5

2

0.5

6

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $12 : 18$

ANSWER $2 : 3$

2. Simplify $30 : 45 : 75$

ANSWER $2 : 3 : 5$

3. Simplify $50\text{ cm} : 2\text{ m}$

ANSWER $1 : 4$

4. Share 120 in $2 : 3$

ANSWER 48 and 72

5. Share 210 in $3 : 4$

ANSWER 90 and 120

6. 5 pens cost 15 000: one pen

ANSWER 3000

7. So 8 pens cost

ANSWER 24 000

Medium · the standard question

7 PROBLEMS

1. Share 300 in 1 : 2 : 3

ANSWER 50, 100, 150

2. A recipe for 4 needs 600 g: for 6

ANSWER 900 g

3. 3 workers lay 90 m a day: 7 workers

ANSWER 210 m

4. 4 workers take 6 days: 8 workers

ANSWER 3 days

5. A map at 1 : 100 000: 7 cm on it

ANSWER 7 km

6. Simplify $\frac{1}{2} : \frac{1}{3}$

ANSWER 3 : 2

7. Cost against number of pens is

ANSWER direct proportion

Hard · stretch and reason

7 PROBLEMS

1. Share 480 in 3 : 5 : 8

ANSWER 90, 150, 240

2. 6 taps fill a tank in 4 hours: 8 taps

ANSWER 3 hours

3. Two shares in ratio 3 : 5 differ by 40: the total

ANSWER 160

4. A map at 1 : 25 000: 3 km on the ground

ANSWER 12 cm

5. Is a child's height directly proportional to age?

ANSWER No — growth is not steady

6. If y is inversely proportional to x and $y = 12$ when $x = 2$, find y when $x = 8$

ANSWER 3

7. A recipe for 4 scaled to 10 multiplies by

ANSWER 2.5

07 Homework

Homework — 5 tasks

Convert to the same units before simplifying any ratio.

1. Simplify 24 : 36, 75 cm : 3 m and 20 : 30 : 50.
2. Share 360 in the ratio 4 : 5.
3. 7 books cost 49 000 sum. Find the cost of 11.
4. 5 workers take 12 days. How long would 10 take?
5. A map at 1 : 50 000 shows two towns 9 cm apart. Find the real distance.